

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0713

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

**Annexure A
Scenario-Based**

Code of Conduct:

I am **Y. Varun Kumar Reddy (192424230)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **Y. Varun Kumar Reddy**

ANSWERS TO SCENARIO-BASED QUESTIONS

Question 1: Smart Campus – Subnetting 192.168.100.0/24

Understanding of the Scenario:

A newly developed Smart Campus has been assigned the IPv4 network 192.168.100.0/24, which provides a total of 256 IP addresses ($2^8 = 256$, because a /24 network leaves 8 bits for host addressing). This address block needs to be divided into three separate subnets for the School of Computing, School of Electronics, and School of Mechanical Engineering. Each department has a different number of devices, so different subnet masks have been assigned: /25, /26, and /27. The subnets should be allocated in descending order of size to ensure efficient utilization of the address space and to minimize fragmentation.

Design Requirements

1. Each subnet must lie completely within the 192.168.100.0/24 network.
2. No two subnets should overlap.
3. The subnet with the highest host requirement should be allocated first.
4. The remaining unallocated address space should be preserved for future expansion.

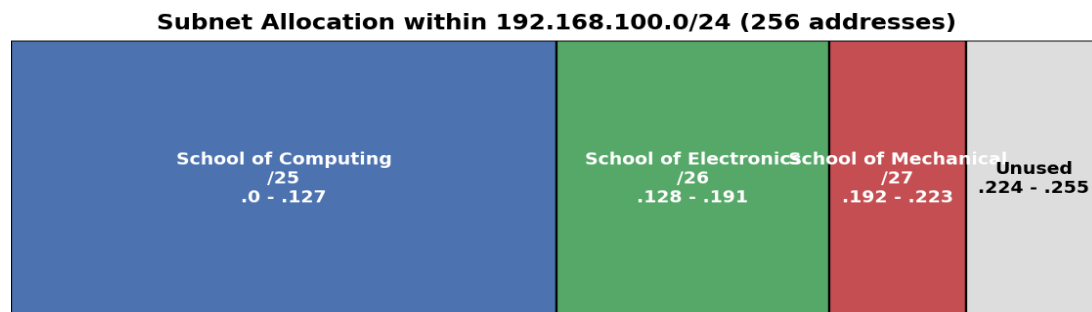
Subnet Mask Breakdown (Binary View):

/25 → 11111111.11111111.11111111.10000000 → Block size = $2^{(32-25)} = 2^7 =$
128 addresses

/26 → 11111111.11111111.11111111.11000000 → Block size = $2^{(32-26)} = 2^6 =$
64 addresses

/27 → 11111111.11111111.11111111.11100000 → Block size = $2^{(32-27)} = 2^5$
= 32 addresses

Allocation Diagram:



1. Network Address and Broadcast Address (Application of Concepts):

School of Computing (192.168.100.0/25): starts at .0, occupies 128 addresses →
Network Address = 192.168.100.0, Broadcast Address = 192.168.100.127.

School of Electronics (192.168.100.128/26): starts immediately after Computing ends, occupies 64 addresses → Network Address = 192.168.100.128, Broadcast Address = 192.168.100.191.

School of Mechanical Engineering (192.168.100.192/27): starts after Electronics ends, occupies 32 addresses → Network Address = 192.168.100.192, Broadcast Address = 192.168.100.223.

School of Computing

- **Subnet:** 192.168.100.0/25
- **Network Address:** 192.168.100.0
- **Broadcast Address:** 192.168.100.127

School of Electronics

- **Subnet:** 192.168.100.128/26
- **Network Address:** 192.168.100.128
- **Broadcast Address:** 192.168.100.191

School of Mechanical Engineering

- **Subnet:** 192.168.100.192/27
- **Network Address:** 192.168.100.192
- **Broadcast Address:** 192.168.100.223

2. Valid Host IP Address Range:

School of Computing: 192.168.100.1 – 192.168.100.126 (excluding network .0 and broadcast .127).

School of Electronics: 192.168.100.129 – 192.168.100.190 (excluding network .128 and broadcast .191).

School of Mechanical Engineering: 192.168.100.193 – 192.168.100.222 (excluding network .192 and broadcast .223).

School of Computing

- **Host Range:** 192.168.100.1 – 192.168.100.126

School of Electronics

- **Host Range:** 192.168.100.129 – 192.168.100.190

School of Mechanical Engineering

- **Host Range:** 192.168.100.193 – 192.168.100.222

3. Number of Usable Host Addresses:

School of Computing (/25): $2^7 - 2 = 126$ usable hosts.

School of Electronics (/26): $2^6 - 2 = 62$ usable hosts.

School of Mechanical Engineering (/27): $2^5 - 2 = 30$ usable hosts.

(Two addresses are always subtracted per subnet — one for the network address and one for the broadcast address — because neither can be assigned to a host.)

4. Remaining Unused IP Address Range (Decision Making):

Total addresses consumed by the three departments = $128 + 64 + 32 = 224$ out of 256. The remaining unused block is 192.168.100.224 – 192.168.100.255, which is exactly a /27-sized block of 32 addresses. This is reserved for future departments or expansion, following good design practice of not exhausting the entire allocated block immediately.

Question 2: Multinational Software Company – Subnetting 10.10.0.0/16

Understanding of the Scenario:

A Multinational Software Company has been assigned the private IPv4 network 10.10.0.0/16, which provides a total of 65,536 IP addresses ($2^{16} = 65,536$, as 16 bits are available for host addresses). The company needs to divide this network into four subnets for the Software Development, Quality Assurance, Human Resources, and Executive Management departments. The subnet masks /24, /25, /26, and /27 have already been decided based on the number of devices required in each department. The subnets should be allocated sequentially from the largest to the smallest to ensure efficient address utilization and easy future expansion.

Design Requirements

1. All subnets must remain within the 10.10.0.0/16 network.
2. The allocated subnets should not overlap.
3. Larger subnet blocks should be assigned before smaller ones.
4. Any unused address space should be identified and reserved for future growth.

Subnet Mask Breakdown (Binary View):

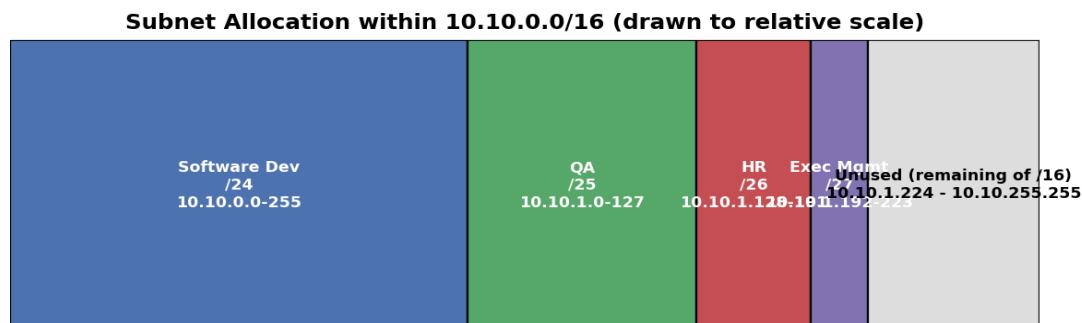
/24 → block size = $2^{(32-24)} = 2^8 = 256$ addresses

/25 → block size = $2^{(32-25)} = 2^7 = 128$ addresses

/26 → block size = $2^{(32-26)} = 2^6 = 64$ addresses

/27 → block size = $2^{(32-27)} = 2^5 = 32$ addresses

Allocation Diagram:



1. Network Address and Broadcast Address:

Software Development (10.10.0.0/24): Network Address = 10.10.0.0, Broadcast Address = 10.10.0.255.

Quality Assurance (10.10.1.0/25): Network Address = 10.10.1.0, Broadcast Address = 10.10.1.127.

Human Resources (10.10.1.128/26): Network Address = 10.10.1.128, Broadcast Address = 10.10.1.191.

Executive Management (10.10.1.192/27): Network Address = 10.10.1.192, Broadcast Address = 10.10.1.223.

2. Valid Host IP Address Range:

Software Development: 10.10.0.1 – 10.10.0.254.

Quality Assurance: 10.10.1.1 – 10.10.1.126.

Human Resources: 10.10.1.129 – 10.10.1.190.

Executive Management: 10.10.1.193 – 10.10.1.222.

3. Number of Usable Hosts:

Software Development (/24): $2^8 - 2 = 254$ usable hosts.

Quality Assurance (/25): $2^7 - 2 = 126$ usable hosts.

Human Resources (/26): $2^6 - 2 = 62$ usable hosts.

Executive Management (/27): $2^5 - 2 = 30$ usable hosts.

4. Unused IP Address Range Remaining (Decision Making):

The four departments together consume $256 + 128 + 64 + 32 = 480$ addresses out of the 65,536 available in the /16 block. The remaining unused range is 10.10.1.224 – 10.10.255.255, i.e. the vast majority of the /16 space, which is kept available for

future departments, additional offices, or network growth without needing to re-plan the existing allocation.

Question 3: Smart Hospital – IPv4 to IPv6 Migration

A Smart Hospital is upgrading its network from IPv4 to IPv6 to accommodate the rapidly increasing number of connected medical equipment, smart sensors, patient monitoring systems, and IoT devices. Since IPv4 offers a limited address space of approximately 4.3 billion addresses, it is no longer sufficient for large-scale healthcare environments. The hospital has been assigned the IPv6 network 2001:0db8:abcd:0000:0000:0000:0000:0000/64, which is the standard subnet size used for IPv6 local area networks (LANs).

Design Considerations

1. The IPv6 address should be compressed according to the standard IPv6 compression rules.
2. The compressed address should also be expanded back to its original 128-bit format.
3. The /64 prefix must be interpreted correctly by separating the network portion from the interface identifier.
4. The total number of available addresses in a /64 subnet should be determined to highlight IPv6 scalability.

Key Issues Identified:

(1) Correctly compressing a long IPv6 address using the two official compression rules. (2) Being able to reverse the process and expand a compressed address back to its full 128-bit hexadecimal form. (3) Understanding that in a /64 prefix, exactly the first 64 bits identify the network and the last 64 bits identify the interface (host). (4) Calculating the enormous address space a /64 subnet provides.

1. Compressed Notation:

Applying IPv6 compression rules — (a) leading zeros within each 16-bit group can be omitted, and (b) one single contiguous run of all-zero groups can be replaced with '::' — the address compresses to: 2001:db8:abcd::/64

2. Expanded (Full Hexadecimal) Representation:

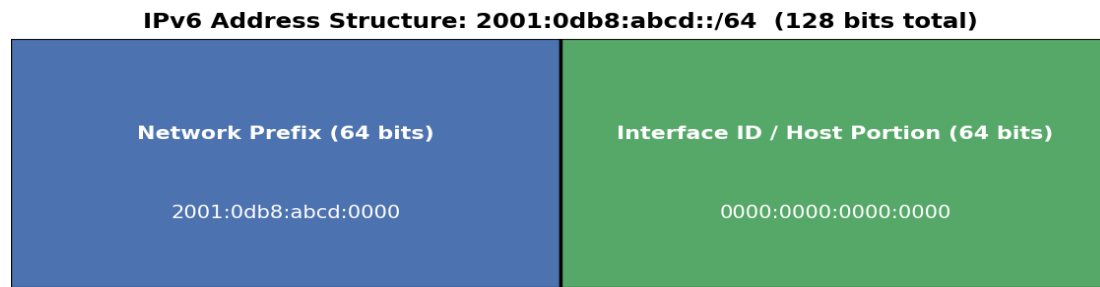
Reversing the compression, each group is restored to its full 4-hex-digit form: 2001:0db8:abcd:0000:0000:0000:0000:0000

3. Network Prefix using the /64 Prefix Length:

A /64 prefix means the first 64 bits (the first four groups) form the network portion, and the remaining 64 bits (last four groups) form the interface identifier. Network

Prefix = 2001:0db8:abcd:0000 (equivalently 2001:db8:abcd::/64).

IPv6 Address Structure Diagram:



4. Total Number of Host Addresses Supported within the Subnet:

Host bits available = 128 total bits – 64 network bits = 64 bits. Total addresses = $2^{64} = 18,446,744,073,709,551,616$ addresses.

5. Total Number of Possible Host Addresses Available within this Network (Decision Making):

Unlike IPv4, IPv6 does not reserve a separate broadcast address, and the all-zero/all-one host addresses within a subnet are not disallowed in the same restrictive way — so effectively the entire 2^{64} (18,446,744,073,709,551,616) address space within the /64 subnet is available for assigning to medical devices, IoT sensors, and other hospital equipment. This provides more than enough headroom for the hospital to expand its connected-device footprint indefinitely without ever needing to re-subnet.

Question 4: ISP Core Router – Routing Table Lookup

Understanding of the Scenario:

An Internet Service Provider (ISP) uses a core router to connect multiple customer networks. The router stores a routing table containing information about all directly connected networks. In this scenario, the routing table includes the following subnets:

- **192.168.1.0/24**
- **192.168.2.0/24**
- **192.168.3.0/24**

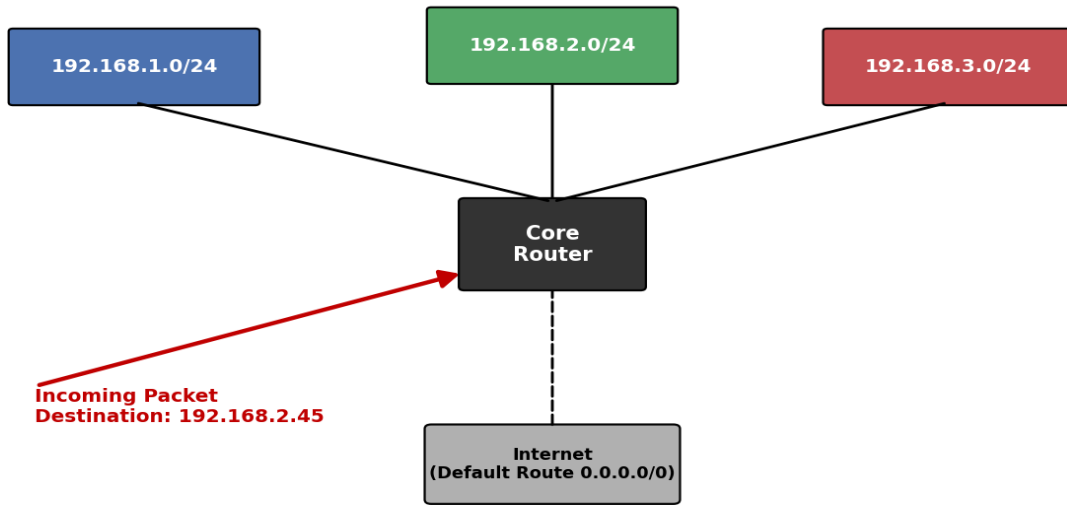
When a packet with the destination IP address 192.168.2.45 reaches the router, it must examine its routing table and determine the correct outgoing interface. The router performs this decision using the Longest Prefix Match (LPM) algorithm, which selects the route with the most specific matching network prefix.

Key Issues Identified:

(1) Correctly identifying which /24 subnet the destination address 192.168.2.45 belongs to. (2) Applying longest-prefix-match logic when multiple entries could partially resemble the destination. (3) Understanding what happens if no route matches — the role of the default route.

Network Topology and Packet Flow:

ISP Core Router - Routing Table Lookup for 192.168.2.45



1. Destination Subnet:

192.168.2.45 lies within the range 192.168.2.0 – 192.168.2.255, so it belongs to subnet 192.168.2.0/24.

2. Matching Routing Table Entry (Application of Concepts):

Comparing the destination address against all three table entries using longest prefix match, only 192.168.2.0/24 matches, since the first 24 bits of 192.168.2.45 (192.168.2) exactly match this entry's network portion.

3. Next-Hop Network/Interface (Decision Making):

Because 192.168.2.0/24 is a directly connected subnet on the router, the packet is forwarded out of the router interface that is physically/logically configured with an IP address belonging to 192.168.2.0/24 (i.e., delivered directly to the destination network rather than to another next-hop router).

4. Default Route (if no match is found):

If the destination address did not match any of the three specific entries, the router would forward the packet using its default route, written as 0.0.0.0/0, which typically points toward the ISP's upstream gateway or the wider internet, ensuring that traffic to unknown networks is still forwarded onward rather than dropped.

Question 5: Two Office Departments via Layer-3 Router

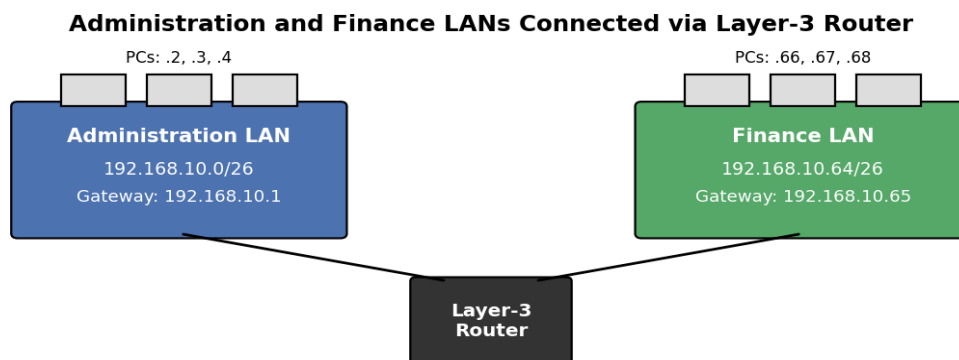
Understanding of the Scenario:

A company has two departments — Administration and Finance — connected through a Layer-3 router, which allows inter-department (inter-VLAN/inter-subnet) communication. Each department has been given a /26 subnet from the 192.168.10.0 block: Administration LAN = 192.168.10.0/26 and Finance LAN = 192.168.10.64/26. The router interfaces act as the default gateway for each LAN.

Key Issues Identified:

(1) Calculating the network and broadcast address for each /26 subnet. (2) Identifying valid host ranges so IP addresses can be assigned without conflict. (3) Confirming that the given gateway addresses (192.168.10.1 and 192.168.10.65) are valid, in-range host addresses for their respective subnets. (4) Making sure suggested computer IP addresses fall inside the valid host range and do not clash with the gateway address.

Network Topology Diagram:



1. Network Address and Broadcast Address:

Administration LAN (192.168.10.0/26): Network Address = 192.168.10.0, Broadcast Address = 192.168.10.63.

Finance LAN (192.168.10.64/26): Network Address = 192.168.10.64, Broadcast Address = 192.168.10.127.

2. Valid Host IP Address Range:

Administration LAN: 192.168.10.1 – 192.168.10.62 (62 usable hosts, since $2^6 - 2 = 62$).

Finance LAN: 192.168.10.65 – 192.168.10.126 (62 usable hosts).

3. Suggested IP Addresses for Three Computers in Each LAN (Application of Concepts):

Administration LAN: PC1 = 192.168.10.2, PC2 = 192.168.10.3, PC3 = 192.168.10.4 (all within the valid range and distinct from the gateway address .1).

Finance LAN: PC1 = 192.168.10.66, PC2 = 192.168.10.67, PC3 = 192.168.10.68 (all within the valid range and distinct from the gateway address .65).

4. Default Gateway Each LAN Should Use (Decision Making):

Hosts in the Administration LAN should be configured with default gateway 192.168.10.1, which is the router's interface IP address for that subnet.

Hosts in the Finance LAN should be configured with default gateway 192.168.10.65, which is the router's interface IP address for that subnet.

Both gateway addresses are correctly chosen — they fall inside their respective valid host ranges and are reserved from being assigned to end-user computers, so all inter-department and internet-bound traffic is properly routed through the Layer-3 router.

RUBRICS FOR CALCULATION

Scenario Based Assessment

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand